

Evanns Morales-Cuadrado

Robotics PhD Candidate | Fulbright Specialist | Georgia Institute of Technology

U.S. Citizen

CONTACT

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SUMMARY

Robotics PhD candidate at Georgia Tech specializing in safe autonomy, reinforcement learning, and real-time control. Built a kinetic-intercept benchmark suite for the U.S. government at Sandia National Laboratories and leads a Lactec collaboration bringing \$20K to Georgia Tech.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

EDUCATION

Georgia Institute of Technology
Ph.D. in Robotics
Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington
Honors Bachelor of Science in Electrical Engineering
Minor in Mathematics
Minor in Physics
Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

2022-Present

2018-2022

WORK EXPERIENCE

Year-Round Graduate Research Intern | AI for Autonomy
Sandia National Laboratories

May 2026 - Present
Albuquerque, NM

- Applied reinforcement learning to autonomy problems in support of national security
- Designed and deployed robotics and automation infrastructure that enables multi-robot experiments
- Built C#/.NET networking frameworks and execution scripts that coordinate communication and task execution across multi-robot systems, now used team-wide

Graduate Research Assistant | FACTS Lab
Georgia Institute of Technology

August 2022 - Present
Atlanta, GA

- Advised by Dr. Samuel Coogan
- Developed a lightweight tracking control law that beat PX4's native tracker in aggressive quadrotor flight and matched NMPC's tracking accuracy at far lower computation time and CPU energy
- Developed safe planning and control methods that combine reachability analysis, formal methods, optimization, and learning to generate fast, provably safe trajectories under unknown disturbances
- Built the lab's real-time flight-test stack, enabling repeatable hardware validation

Research Assistant | DN&C and Autonomous Systems Labs

October 2019 - May 2022

University of Texas at Arlington

Arlington, TX

- Advised by Dr. Frank Lewis in the Autonomous Systems Lab and Dr. Yan Wan in the Dynamical Networks and Control Lab
- Contributed to reinforcement-learning quadrotor control research under Dr. Frank Lewis, then ranked 12th in the USA and 23rd worldwide
- Completed UTA Honors College thesis on lightweight Newton-Raphson tracking control for computationally constrained aerial systems
- Conducted original research on learning-based minimum-time and minimum-energy multi-vehicle path planning using ROS and OpenCV

**SELECTED
PUBLICATIONS**

E. Morales-Cuadrado, L. Chung, S. Kousik, and S. Coogan, “RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 MECC-ALDSC. (*accepted to both*) [preprint available](#).

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (*under revision*) [preprint available](#).

E. Morales-Cuadrado, L. Baird, Y. Wardi, and S. Coogan, “Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method.” [IEEE Transactions on Control Systems Technology \(TCST\)](#), 2026. [preprint available](#).

**OPEN-SOURCE
CONTRIBUTIONS**

- **PX4 Autopilot**: Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4**: Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

**SERVICE &
LEADERSHIP**

Co-Founder and President of Puerto Rican Student Association

August 2024 - Present

Georgia Institute of Technology

Atlanta, GA

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

**HONORS AND
AWARDS**

Fulbright Specialist

August 2026

Peer-review-selected to the U.S. Department of State’s Fulbright Specialist Roster for a three-year term

Top-3 Finalist at a Deep Learning Research Symposium

November 2023

Professor-sponsored award for novel research in deep learning and robotics

Goizueta Fellowship at Georgia Tech

August 2022

Highly selective fellowship for graduate students of Latin-American descent

Summa Cum Laude Honors at University of Texas at Arlington

May 2022

Exceptional academic distinction in top GPA tier for graduation honors

Electrical Engineering Honors Scholar at University of Texas at Arlington

December 2021

Inaugural member of the electrical engineering honors cohort

IEEE-HKN International Electrical Engineering Honor Society Membership

August 2021

Selective undergraduate society recognizing outstanding academic achievement and community service